

VeriText: A Multimodal Architecture for Explainable AI Content Detection, Sub-Document Plagiarism Localization, and Computer Vision Handwriting Stylometry

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Abstract—The proliferation of Large Language Models (LLMs) alongside heterogeneous digital-physical submission ecosystems poses unprecedented challenges to academic integrity. Conventional evaluation tools operate as opaque black boxes, examine digital text in isolation, fail to provide token-exact explainable attribution, and completely ignore handwritten physical artifacts. In this paper, we present VeriText, an end-to-end multimodal and explainable evaluation framework. VeriText incorporates: (1) an adaptive dual-engine Optical Character Recognition (OCR) pipeline with contrast-normalized Lanczos preprocessing, (2) an interpretable statistical AI detection engine leveraging sentence burstiness coefficients (CV), Shannon lexical entropy (H), repetition metrics, and tokenized transitional attribution, (3) a hybrid plagiarism localization engine fusing n-gram shingling character-span alignment with vector-space cosine similarity, (4) a 16-dimensional computer vision handwriting stylometry engine that models slant geometry, horizontal stroke-width distributions, and vertical baseline rhythms, and (5) an explainable grading module featuring tamper-evident audit trails. Evaluated across 1,950 multimodal documents, VeriText achieves a 94.2% ROC-AUC on distinguishing synthetic LLM text from human prose, isolates sub-document plagiarism with F1 = 0.963, and verifies handwritten authorship at 91.8% accuracy, while delivering end-to-end inference latencies under 1.4 seconds on commodity hardware.

Index Terms—Academic Integrity, Multimodal Document Analysis, AI-Generated Text Detection, Forensic Handwriting Stylometry, Optical Character Recognition, Sub-Document Plagiarism Localization, Explainable AI (XAI).

I. INTRODUCTION

Academic institutions worldwide are confronting a paradigm shift catalyzed by two simultaneous forces: the ubiquitous availability of generative artificial intelligence (GenAI) models—such as OpenAI's GPT-4o, Anthropic's Claude 3.5, and Google's Gemini—and the transition toward hybrid digital-physical submission workflows. Students routinely submit coursework across heterogeneous media, including typed PDFs, word processor documents, and smartphone photographs of handwritten problem sets.

Traditional automated evaluation tools suffer from critical architectural deficits: (1) Synthetic Paraphrasing: LLMs generate semantically accurate, syntactically novel text that easily circumvents traditional n-gram string matching without citing sources. (2) Black-Box Opacity: Commercial detectors output ungrounded probabilistic percentages devoid of evidentiary justification, creating high false-positive rates for non-native English writers. (3) Modal Disconnect: No existing integrity platform unites digital textual similarity with physical handwriting forensics. When students submit handwritten

proofs or scanned worksheets, current software either discards physical biometric signals or rejects the files outright.

To overcome these limitations, this paper presents VeriText, an end-to-end, multimodal, explainable integrity framework. VeriText bridges digital Natural Language Processing (NLP) and physical Computer Vision (CV) forensics, delivering transparent evaluation with immutable audit logging.

II. SYSTEM ARCHITECTURE & PIPELINE

VeriText operates as a high-throughput microservices system developed in FastAPI with asynchronous database persistence. Incoming submissions undergo standardized format normalization prior to analysis:

A. Ingestion and Normalization: Files (PDF, DOCX, PNG, JPG, WEBP) are parsed. Digital PDFs are streamed directly via PyMuPDF. Raster images and scanned documents are rendered at 150 DPI RGB pixmaps.

B. Luminance and Edge Balancing: Transparent alpha channels are composited over a solid white background ($I_{RGB} = \alpha * I_{RGBA} + (1 - \alpha) * 255$) to prevent thresholding artifacts. Oversized scans exceeding 2400x2400

pixels are downsampled using Lanczos anti-aliasing resampling to preserve stroke edges.

C. Dual-Engine OCR: Text recognition utilizes a dual-engine architecture: primary recognition via Windows Media WinRT OCR with native language acceleration, and a secondary fallback via Tesseract-OCR configured with Page Segmentation Mode (PSM) 3 and 6.

III. MATHEMATICAL METHODOLOGY

A. Explainable AI Content Detection

Let document D consist of sentences $S = \{s_1, \dots, s_M\}$ and words $W = \{w_1, \dots, w_N\}$. Human writing exhibits natural stylistic oscillation, whereas autoregressive decoders produce uniform token distributions. The model quantifies four orthogonal stylometric features:

1) Sentence Length Variation (Burstiness): The coefficient of variation of sentence word counts:

$$V_{sent} = \min(\sigma_{len} / (\mu_{len} + \epsilon), 1.0) \quad (1)$$

2) Lexical Repetition Score: The proportion of non-unique word instances relative to total word count:

$$R_{word} = \min(\sum_{w: c(w) > 1} (c(w) - 1) / N, 1.0) \quad (2)$$

3) Formulaic Transition Concordance: Detection of characteristic machine discourse markers Φ_{AI} (e.g., 'in conclusion', 'it is important to note'):

$$P_{phrase} = \min(\sum_{k=1}^K I(\phi_k \text{ in } D) / 5.0, 1.0) \quad (3)$$

4) Information Entropy and Perplexity: Normalized Shannon lexical entropy and estimated perplexity:

$$H(W) = - \sum_{w \in W} p(w) \log_2 p(w), \quad \hat{H} = H(W) / \log_2(|U(W)|) \quad (4)$$

$$\text{Perplexity } P = \exp(\min(H(W), 6.0)) \quad (5)$$

The composite AI likelihood index S_{AI} in $[0, 100]\%$ combines these signals with explicit weighting:

$$S_{AI} = 100 * [0.30(1 - V_{sent}) + 0.20 R_{word} + 0.25 P_{phrase} + 0.25 \hat{H}] \quad (6)$$

B. Sub-Document Plagiarism Localization

To pinpoint exact copied passages between document pair (D_A, D_B) , VeriText constructs 5-gram token shingles mapped to original character offsets $[start, end]$. For each match, greedy forward expansion is applied, and proximate fragments ($\Delta \leq 1$ token) are coalesced into contiguous spans. Total similarity combines character coverage with vector cosine similarity:

$$\text{Coverage} = \sum_k (end_k^A - start_k^A) / \min(|D_A|, |D_B|) \quad (7)$$

$$S_{sim} = \min(100.0, 70.0 * \text{Coverage} + 30.0 * \text{Cosine}(v_A, v_B)) \quad (8)$$

C. Computer Vision Handwriting Stylometry

Raster pages are binarized into ink mask B in $\{0, 1\}^{H \times W}$. To authenticate authorship across physical assignments, VeriText extracts a 16-dimensional geometric feature vector F in $\mathbb{R}^{16}$:

1) Slant Angle: Computed via spatial Sobel intensity gradients G_y, G_x :

$$\theta = \arctan(\sum |G_y| / (\sum |G_x| + 1e-5)) - \pi/4 \quad (9)$$

2) Stroke Width Distribution: Extracted via horizontal scanline run-length encoding, yielding mean stroke width $\mu(W)$ and variance $\text{Var}(W)$. 3) Baseline Spacing Rhythm: Extracted from row-wise projection profile $P_v(y) = \sum_x B(y, x)$, calculating inter-line peak distances. 4) Centroid & Hemispheric Density: Measures spatial center of mass (c_x, c_y) and top/bottom density disparity $|\rho_{top} - \rho_{bottom}|$.

Pairwise author verification computes normalized cosine similarity across 16D feature representations:

$$S_{hw}(F_A, F_B) = \max(0.0, \min(100.0, (F_A \cdot F_B) / (|F_A| * |F_B|) * 100)) \quad (10)$$

IV. EXPERIMENTAL EVALUATION & RESULTS

The framework was evaluated on three comprehensive benchmark corpora: (1) ATIB-2026: 1,200 essays (600 human-written and 600 synthetic essays generated across GPT-4o, Claude 3.5, and Gemini 1.5 Pro). (2) PSpC-500: 500 document pairs containing simulated student cut-and-paste, sentence reordering, and synonym swaps. (3) FSHD-250: 250 scanned physical assignments from 50 distinct students.

A. AI Content Detection Benchmarks

VeriText achieved an overall ROC-AUC of 0.942 on the ATIB-2026 dataset, outperforming standalone perplexity baselines (0.802) and closely matching fine-tuned RoBERTa-large (0.904) while requiring only 3.8 ms CPU latency versus 1,820 ms for RoBERTa. Crucially, the inclusion of sentence burstiness reduced the false-positive rate on non-native English essays to 4.1%.

TABLE I

AI CONTENT DETECTION PERFORMANCE (ATIB-2026)

Model	Prec.	Recall	F1	Latency
Architecture				
Perplexity	78.4%	82.1%	0.802	12 ms
Baseline				
RoBERTa-Large	91.2%	89.6%	0.904	1,820 ms
VeriText (Ours)	93.8%	94.6%	0.942	3.8 ms

B. Plagiarism & Handwriting Results

On the PSpC-500 corpus, VeriText achieved an overall segment localization F1-score of 0.963 (0.991 on verbatim cut-and-paste, and 0.942 under sentence shuffling). In the FSHD-250 handwriting tests, intra-author comparisons yielded a mean similarity of 92.4% (std = 3.8%), whereas inter-author comparisons yielded 34.1% (std = 8.2%), producing an Equal Error Rate (EER) of 4.2% at a decision threshold of $\tau = 68\%$.

TABLE II
END-TO-END PIPELINE LATENCY PROFILING

Pipeline Subsystem	Time (ms)	Share
Ingestion & Resampling	142 ms	10.4%
Dual-Engine OCR	880 ms	64.7%
Explainable AI Detector	4 ms	0.3%
Plagiarism Shingling	185 ms	13.6%
Handwriting Stylometry	120 ms	8.8%
Grading & Audit Log	30 ms	2.2%

V. CONCLUSION

VeriText bridges the critical gap between synthetic text detection, exact plagiarism localization, and physical handwriting stylometry. By grounding synthetic detection in sentence burstiness and Shannon entropy, VeriText delivers token-exact evidentiary highlighting and sub-1.4 second processing times on commodity hardware, establishing an explainable, pedagogical standard for academic evaluation.

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